

Graph-Based Fragment Detection and Precursor Assembly

A 2D visual guide to the implemented single-step recommendation workflow

From inventory to precursor recommendations

Recognize useful molecular pieces, then combine complementary pieces around a desired product.

1. Fragment detection

Find coherent pieces of an available compound that are conserved in the desired product.

mapped pieces

2. Precursor assembly

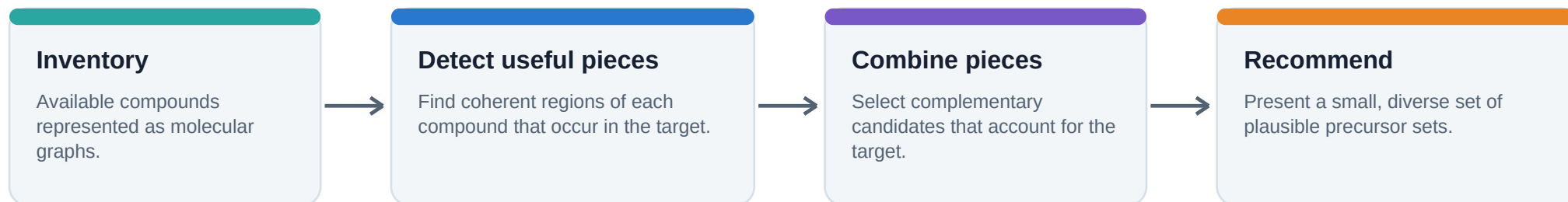
Combine complementary pieces into complete, explainable product construction patterns.

All colored molecular regions in this report come from computed atom mappings.

The chemistry drawings are 2D for clarity; routine C-H labels are omitted.

Objective and general approach

Recommend available starting materials whose conserved pieces can account for a desired product



What the workflow is trying to preserve

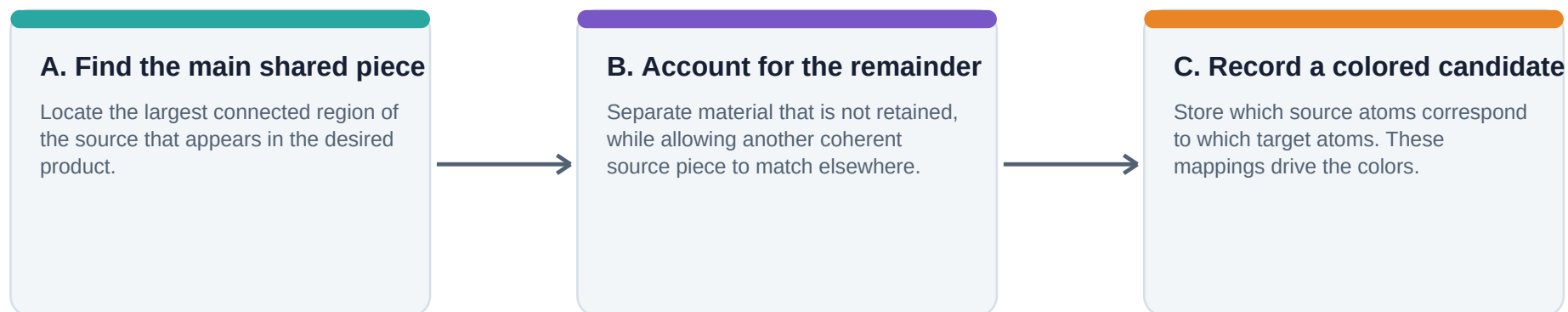
- Large, recognizable molecular pieces are preferred over scattered atom-by-atom coincidences.
- Each product atom is assigned to one proposed source, giving the recommendation clear provenance.
- Different ways of dividing the target remain visible as distinct construction patterns.
- Every 2D color comes from the computed atom correspondence rather than manual annotation.

Important scope

This is a structural recommendation system. It proposes coherent atom sources and implied bond edits; it does not by itself prove kinetic feasibility, select catalysts, or guarantee a one-step laboratory reaction.

Stage 1: detect useful molecular pieces

Compare each available compound with the target and retain coherent shared regions



Graph interpretation



The important idea is fragment-level conservation. A source can contribute more than one coherent piece, but atoms that do not belong in the product are not forced into arbitrary product positions.

Example 1: one source contributes multiple fragments

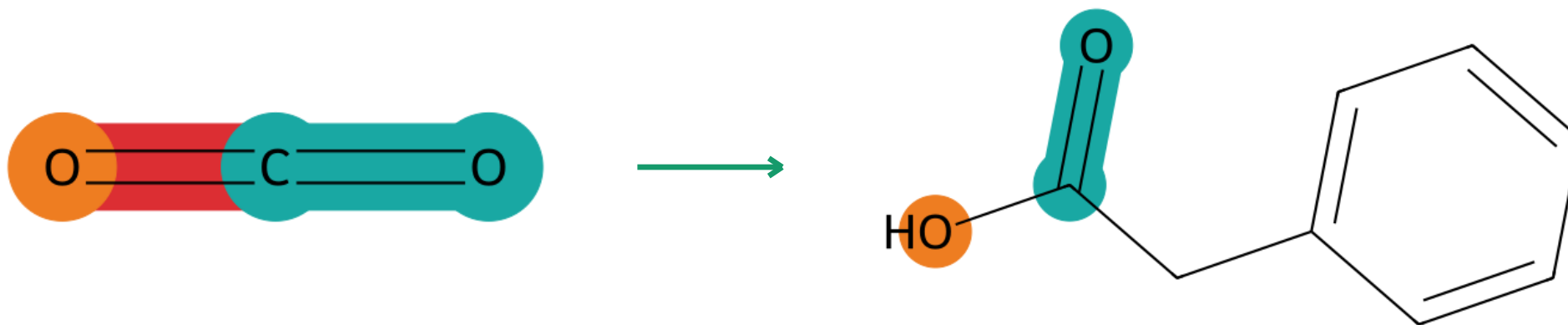
CO₂ is recognized within the carboxyl group of phenylacetic acid

Source: carbon dioxide

The carbonyl piece is recognized first. The second oxygen is then retained as another useful piece of the same source.

Target: phenylacetic acid

Only the colored carboxyl atoms are owned by this candidate. The aromatic side of P remains available to other sources.



● initial retained fragment

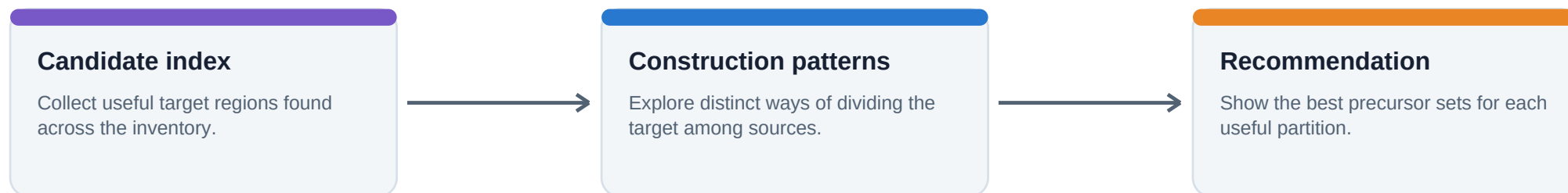
● additional target-owned fragment

What the colors show

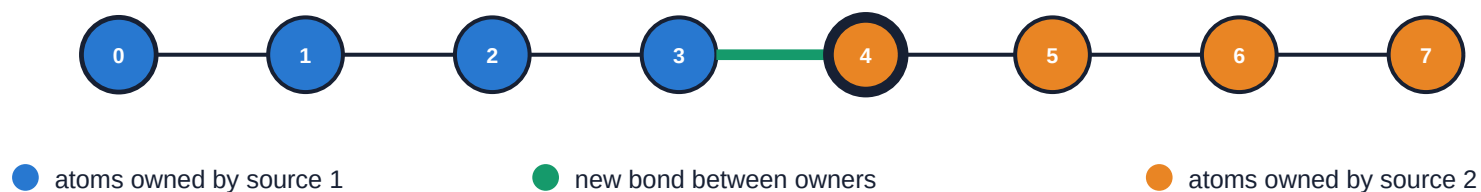
The cyan carbonyl piece is the largest connected match. The orange oxygen is then recognized as another valid part of the target carboxyl group. Together, all three CO₂ atoms receive product positions; the rest of phenylacetic acid remains available for other precursors.

Stage 2: assemble a complete product explanation

Combine complementary pieces so every target atom has one proposed source



A target atom has exactly one owner

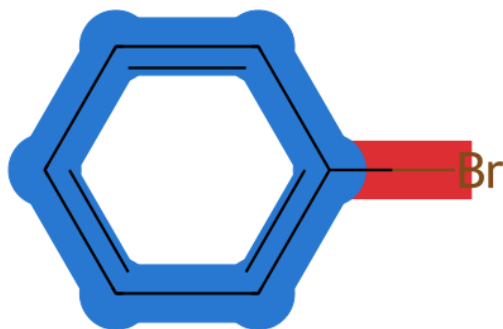


- Selected pieces do not overlap: each product atom has one proposed source.
- Together, the selected pieces explain the complete product graph.
- A bond joining two colors is a proposed connection between source-derived modules.
- The result is a recommendation for review, not a claim that reaction conditions are known.

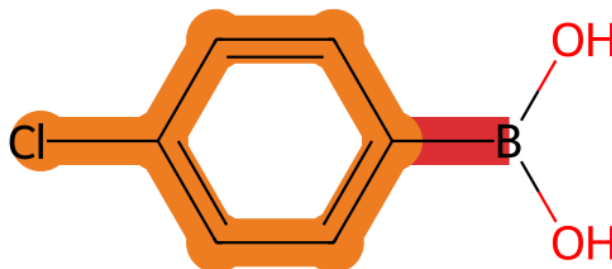
Example 2: clean two-source assembly

Inventory compounds: bromobenzene and 4-chlorophenylboronic acid

Bromobenzene - inventory barcode C0042016



4-Chlorophenylboronic acid - barcode C0115718



4-Chlorobiphenyl target



● blue source ownership

● orange source ownership

● red source bond cut

● green product bond formed

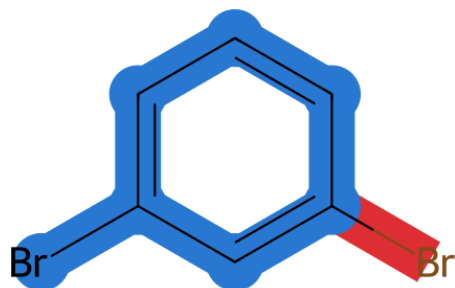
Computed result

- Each aryl ring remains a coherent colored module from its original inventory compound.
- Bromine and the boronic-acid group are left uncolored because they are not retained in the target.
- The two colors cover the complete product without competing for the same atom.
- The green bond is the proposed connection between the two source-derived modules.

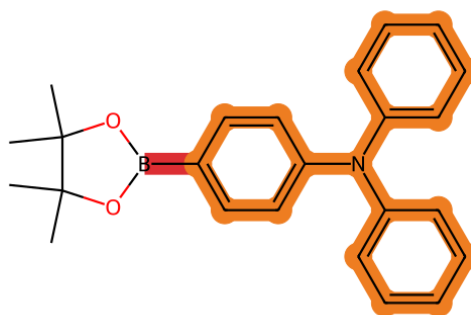
Example 3: a larger inventory-derived target

The algorithm colors atom ownership, not a hand-selected reaction template

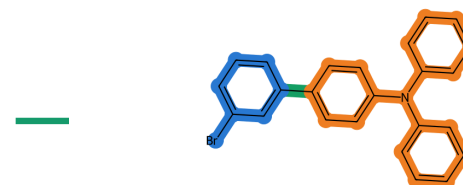
1,3-Dibromobenzene - C0113869



Triarylamine BPin - C0129716



Assembled target



● retained smaller aryl module

● retained triarylamine module

● new inter-module bond

What this example demonstrates

- The detector keeps the large triarylamine aryl system coherent instead of decomposing it into atom-wise matches.
- The unused bromine and the BPin unit remain residual source material and are not colored as product ownership.
- Together, the two colored candidates explain the complete target without overlapping ownership.
- The same general approach works for both a simple coupling and this substantially larger structure.

Inventory integration and interpretation

The structure bank provides candidates; graph matching supplies atom-level evidence

3,059 containers

Physical inventory rows retained with barcode and location metadata.

2,769 resolved

Containers with a PubChem structure that parses in RDKit.

1,919 structures

Unique stereochemical SMILES in the deduplicated search bank.

290 unresolved

Not-found or ambiguous identifiers remain excluded and auditable.

Recommended use

- Compare every resolved inventory structure with the desired target.
- Collect compounds that contribute substantial coherent pieces.
- Combine complementary pieces into several distinct construction patterns.
- Review the leading recommendations with stereochemical and chemical knowledge.

Reading the colors

Blue, orange, and purple identify source ownership. Red marks a source boundary bond cut. Green marks a product bond connecting different owners. Uncolored atoms remain in a source residual or are not supplied by that candidate. Hydrogens participate explicitly in the computation, although routine C-H labels are hidden in the skeletal drawings for legibility.

Recommendations favor a small number of source structures that retain a large fraction of their atoms with few structural changes. This ordering is explainable but is not a substitute for reaction conditions or mechanistic judgment.

How to interpret the result

What the report supports

- A visual explanation of which molecular pieces can be conserved in the target.
- A transparent proposal for which inventory compounds supply each product region.
- A compact set of structurally distinct precursor recommendations.
- A starting point for a chemist to assess reactivity, conditions, selectivity, and availability.

References

1. PubChem PUG REST: <https://pubchem.ncbi.nlm.nih.gov/docs/pug-rest>
2. RDKit documentation: <https://www.rdkit.org/docs/>
3. Project design specification: RETROSYNTHESIS_DESIGN.md in the coordinate_alignment repository.

Figure provenance

Regenerate this PDF with `tools/build_fragment_matching_report.py`. Molecular highlights are recomputed from the current detector and assembly implementation each time.